

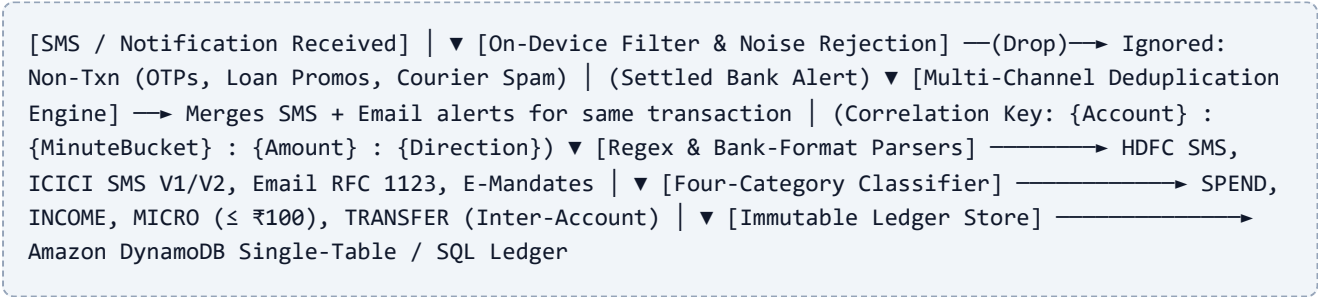
Simplify Money — Product Track Flow Teardown

Author: Dipanshu Raj | **Feature:** "Track" Flow (Automated Financial Ledger Synchronization) | **Repository:** github.com/RjDipanshu/ledger-sync-seed

1. Executive Summary

Simplify Money's core value proposition answers the fundamental question: *"Where did my money go?"* without forcing users to manually enter transactions. The **"Track"** flow serves as the operational engine of this experience. It ingests semi-structured notification streams (SMS, notification listener, bank alert emails), normalizes them into immutable transactional ledger records, rolls up micro-expenses, and maintains running balances across multiple accounts.

2. End-to-End Architecture & Ingestion Journey



Stage-by-Stage Breakdown:

- **Stage 1: Permission Priming & User Trust:** Financial tracking requires Android SMS permissions. The app primes users with clear visual cards explaining that banking OTPs and personal messages are strictly ignored.
- **Stage 2: Filter & Ingestion:** Out of 522 raw messages in Corpus-A, 41 non-transactional messages (promotional loan offers, ATM OTPs, delivery updates) are safely filtered without entering the ledger.
- **Stage 3: Multi-Message Deduplication:** Banks issue multiple alerts (SMS, Email, UPI app push) for a single payment. Deduplication merges these using `{account_last4}:{minute_bucket}:{amount}:{direction}` , consolidating `source_message_ids` .
- **Stage 4: Categorization:**
 - `MICRO` : UPI debits $\leq ₹100$ are grouped as rollups to prevent visual clutter in the main feed.
 - `TRANSFER` : Intra-user money movements between known accounts are isolated to prevent inflating spend and income.
 - `SPEND` & `INCOME` : Real outflows and inflows accurately reflecting net worth shifts.

3. Identified Friction Points & Engineering Resolutions

Friction Point	Observed Manifestation	Engineering Resolution
Whole-Rupee Regex Trap (INC-2026-09-11)	Whole-rupee alerts (Rs. 5) failed two-decimal regex and erroneously captured Avl Bal: Rs.92,213.10 .	Updated amount regex to (?:Rs\. INR)\s*([0-9,]+(?:\.[0-9]{2})?) with strict trailing punctuation lookaheads.
Silent Bank SMS Drops	Bank network outage or dropped SMS leads to unaccounted balance drops (e.g. ₹7,500 gap on Account 4821).	Compare consecutive stated balances to identify gaps; flag discrepancies in reconciliation.json .
Notification Timing Latency	SMS received minutes or hours late due to roaming or network latency.	Parse transaction timestamp inside message body rather than relying on phone reception time (received_at).

4. Architectural Recommendations for Scale

- 1. **On-Device Hybrid Parsing vs. Cloud Processing:** Parsing bank regex patterns on-device protects user privacy by never sending raw message text or sensitive numbers to cloud servers. Only normalized transaction payloads are synchronized.
- 2. **Reconciliation as a First-Class Feature:** Rather than hiding unexplained discrepancies, surface them cleanly to users: "We noticed a ₹7,500 withdrawal on 29 July not captured via SMS. Would you like to tag this as Cash Withdrawal?"
- 3. **DynamoDB Single-Table Design:** Partitioning transactions by ACC#<last4>#<YYYY-MM> provides guaranteed \$O(1)\$ monthly statement reads with zero table scans. Category summaries are pre-aggregated on a single summary item (ScannedCount = 1, Count = 1).

Conclusion

The Track flow is the foundation of user retention in personal finance. Handling edge cases, multi-message deduplication, and honest balance reconciliation ensures the ledger remains trustworthy and transparent.